

Qatar-1b

R-2539

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190819 · created 2026-08-25T23:57:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458714.8833 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.054
Transit depth (Rp/Rs)^2	3.71 +/- 2.1 [%]
Orbital Inclination (inc)	88.91 +/- 2.78
Airmass coefficient 1 (a1)	0.162 +/- 0.071
Airmass coefficient 2 (a2)	1.07 +/- 0.2
Scatter in the residuals of the lightcurve fit is	32.45 %
Best Comparison Star	None
Optimal Aperture	5.94
Optimal Annulus	11.84
Transit Duration (day)	0.0835 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.054 vs 0.1463 (deviation 0.31σ)
Duration fit vs published	0.0835 d vs 0.0692 d (deviation 0.74σ)
Mid-time O–C	-3.6 min
Depth SNR	1.3
Residual scatter	32.45 %

Light curve

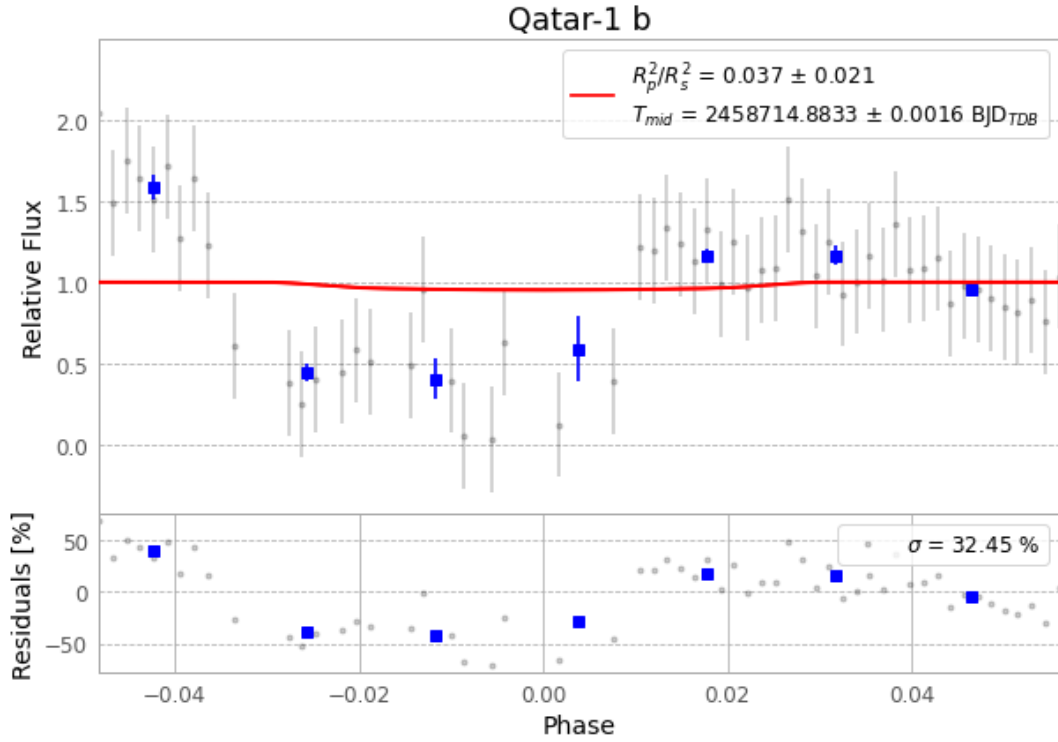


Figure 1 — FinalLightCurve_Qatar-1 b_2019-08-19.png (SHA-256 6911e3c57254a766...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8f2b324e45b586b6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1190819073012.FITS → Qatar-1190819110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190819.log (eb3371f39400...), FinalLightCurve_Qatar-1 b_2019-08-19.png (6911e3c57254...), FinalLightCurve_Qatar-1 b_2019-08-19.csv (a0af5eaa7a58...)

Compute resources

Wall time 195.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:57Z.